MILITARY SPECIFICATION

HOSE, SYNTHETIC RUBBER, WIRE REINFORCED, AND END FITTINGS, REUSABLE, FOR FLEXIBLE HOSE ASSEMBLIES (DOUBLE-WIRE BRAID FOR MEDIUM PRESSURE SIZES -40, -48, and -64)

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers synthetic rubber hoses reinforced by two layers of wire braid in sizes -40, -48, and -64 for medium pressure hydraulic applications.

1.2 Classification. End fittings for hoses covered by this specification shall be sizes -40, -48, and -64 as applicable and shall be of the following types:

Type AI - 400 pounds per square inch (lb/in^2) flange, to hose, straight, tube sizes 2 to 4 inches (see figure 1).
Type AII - 250 lb/in^2 flange, to hose, straight, tube sizes 2 to 4 inches (see figure 2).
Type AIII - 150 lb/in^2 flange, to hose, straight, tube sizes 2 to 4 inches (see figure 3).
Type AIV - 150 lb/in^2 flange, to hose, straight, tube sizes 2 to 4 inches (see figure 4).
Type BI - 400 lb/in^2 flange, to hose, 90-degree elbow, tube sizes 2 to 4 inches (see figure 5).
Type BII - 250 lb/in^2 flange, to hose, 90-degree elbow, tube sizes 2 to 4 inches (see figure 6).
Type BIII - 150 lb/in^2 flange, to hose, 90-degree elbow, tube sizes 2 to 4 inches (see figure 7).
Type BIV - 150 lb/in^2 flange, to hose, 90-degree elbow, tube sizes 2 to 4 inches (see figure 8).
Type C - Split clamp to hose, straight, with female braze tailpiece and clamp assembly, tube sizes 2 to 4 inches (see figures 9, 15, 16).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Type CA - Split clamp to hose, straight, with male braze tailpiece and clamp assembly, tube sizes 2 to 4 inches (see figures 10, 15, 17).
Type D - Split clamp to hose, 90-degree elbow, with female braze tailpiece and clamp assembly, tube sizes 2 to 4 inches (see figures 11, 15, 16).
Type DA - Split clamp to hose, 90-degree elbow, with male braze tailpiece and clamp assembly, tube sizes 2 to 4 inches (see figures 12, 15, 17).
Type E - 90-degree elbow fitting, hose to hose (see figures 13, 15).
Type F - Split clamp fitting, 180-degree elbow, with split clamp assembly, hose to hose (see figures 14, 15).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications. The following specifications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-C-390 - Copper Alloy Castings (Including Cast Bar).
QQ-C-591 - Copper-Silicon, Copper-Zinc-Silicon, and Copper-Nickel-Silicon Alloys: Rod, Wire, Shapes, Forgings, and Flat Products (Flat Wire, Strip, Sheet, Bar and Plate).
QQ-N-281 - Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections.
QQ-N-286 - Nickel-Copper-Aluminum Alloy, Wrought (UNS N05500).

MILITARY

MIL-L-2104 - Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-L-17331 - Lubricating Oil, Steam Turbine and Gear, Moderate Service.
MIL-F-20042 - Flanges, Pipe and Bulkhead Bronze (Silver Brazing).
MIL-H-24135 - Hose, Synthetic Rubber, Wire Reinforced, and End Fittings, Reusable, for Flexible Hose Assemblies, General Specification for.
MIL-B-24480 - Bronze, Nickel-Aluminum (UNS No. C95800) Castings for Seawater Service.

(Copies of specifications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Hose. Hose furnished under this specification shall conform to the requirements of MIL-H-24135 and as specified herein.

3.2 Material.

3.2.1 Hose. The rubber compounds used shall be resistant to petroleum base oils and sea water and shall consist of synthetic rubber or a blend of synthetic rubbers and shall meet the requirements of MIL-H-24135 and as specified herein.

3.2.2 Fittings. Material for fittings shall be as shown on figures 1 through 17.

3.3 Reinforcement. The reinforcement shall consist of two wire braids applied in such a manner as to meet all the requirements of this specification.

3.4 Dimensions and pressures. Hose dimensions and pressures shall be in accordance with table I.

TABLE I. Dash numbers, performance characteristics, and dimensions.

Hose size (dash no.)	Pressure (lb/in^2)		Hose dimensions (inches)						Min bend radius (inches)
	Working (max)	Burst (min)	Id tube		Od cover		Od reinf		
			min	max	min	max	min	max	
-40	1,200	4,800	2.500	2.562	3.031	3.157	2.891	2.985	26.00
-48	900	3,600	3.000	3.062	3.531	3.657	3.391	3.485	30.00
-64	600	2,400	4.000	4.062	4.640	4.765	4.499	4.593	33.00

3.5 Fitting types and sizes. Fitting types and sizes shall be as shown on figures 1 through 17.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance provisions shall be in accordance with MIL-H-24135 and as specified herein.

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4.2 Qualification samples for testing.

4.2.1 Hose. Qualification testing of hose shall consist of hose samples as specified in table II. Samples shall be not less than 15 inches in length and shall be fitted with qualified end fittings.

TABLE II. Numbers of hose samples for qualification testing.

Hose size (dash no.)	Number of hose samples to be provided
-40	4
-64	5

4.2.2 End fittings. Qualification testing of end fittings only shall consist of 18 end fittings of the same type as specified in table III. End fitting samples shall be installed on nine qualified hose samples, in the sizes listed in table II.

TABLE III. Numbers of end fitting samples for qualification testing.

End fitting size (hose dash no.)	Number of end fitting samples to be provided
-40	8
-64	10

4.2.3 Hose and end fittings combined. When a single contractor of both hose and fittings qualifies both hose and fittings, or a hose contractor and a fitting contractor qualifies the hose and fittings in a combined qualification test, the samples shall be provided as specified in table IV.

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TABLE IV. Number of samples for combined qualification testing.

One approved source					
Hose size (dash no.)	No. of hose samples to be provided		End fitting (hose dash no.)	No. of end fittings to be provided	
	Qualifier 2/	Approved 1/ source		Qualifier 2/	Approved 1/ source
-40	2	2	-40	4	4
-64	3	2	-64	5	5

Two or more approved sources						
Hose size (dash no.)	Number of samples to be provided by			End fitting size (dash no.)	Number of end fittings to be provided by		
	Qualifier 2/	Approved 1/ source no. 1	Approved 1/ source no. 2		Qualifier 2/	Approved 1/ source no. 1	Approved 1/ source no. 2
-40	2	1	1	-40	4	2	2
-64	3	1	1	-64	5	3	2

Hose samples shall be not less than 15 inches in length and shall be coupled on one end with the fitting being submitted for qualification and on the other end with previously qualified end fittings.

1/ Approved source - Previously qualified hose and fittings, or hose or fitting.
2/ Qualifier - Hose and fittings, or hose or fitting to be qualified.

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4.2.4 Sequence of tests on sample hose assemblies shall be as specified in table V.

TABLE V. Sequence of qualification tests on sample assemblies.

	Sample numbers								
	Size (dash no.) -40				Size (dash no.) -64				
Test sequence	1	2	3	4	5	6	7	8	9
Proof pressure (see 4.6.1)1/	X	X	X	X	X	X	X	X	X
Stability test (see 4.6.2)1/	X	X	X	X	X	X	X	X	X
Aging test (see 4.6.3)1/		X	X	X		X	X	X	X
Hydraulic fluid circulation (see 4.6.4)1/		X				X			
Impulse test (see 4.6.5)1/				X				X	X
Burst test (see 4.6.6)1/	X				X				
Vacuum test (see 4.6.7)1/			X				X		
Cold temperature flexing (see 4.6.8)1/			X				X		
Adhesion test (see 4.6.9)1/			X				X		

1/ See MIL-H-24135.

4.3 The test fluid shall be petroleum base oil in accordance with MIL-L-17331 or MIL-L-2104 for the aging, hydraulic fluid circulation, and impulse tests.

4.4 The peak pressure for the impulse test shall be 150 percent of maximum working pressure (see table I).

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5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 The packaging requirements shall be in accordance with MIL-H-24135.

6. NOTES

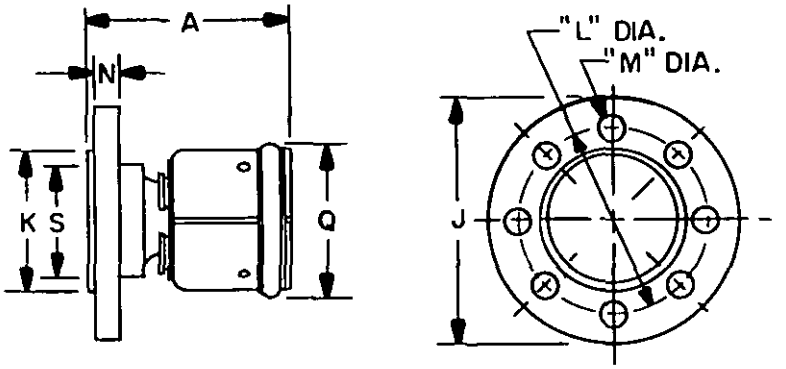
6.1 Intended use. Hose and fittings to this specification are intended for use in shipboard fluid piping systems where pressures do not exceed the rated working pressure of the hose and temperature does not exceed minus 40 to 200°F.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Type of end fittings for hoses (see 1.2).

Preparing activity:
Navy - SH
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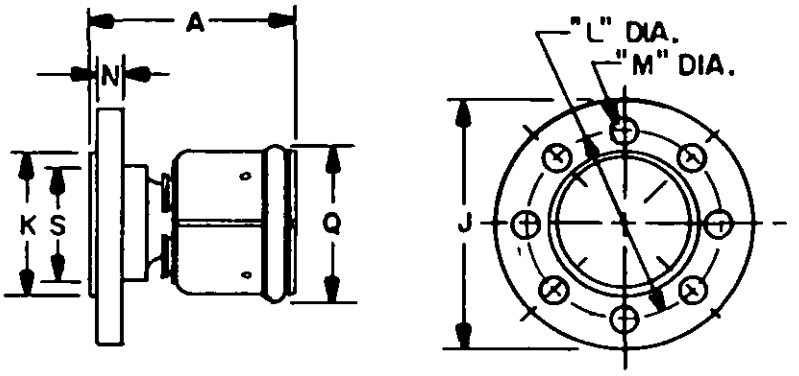
Hose size (dash no.)	Nps	Number of bolt holes	A	J	K	L	M (bolt circle)	N (bolt holes)	Q	S
-40	2	7	5.16	6.50	3.28	5.19	0.69	0.81	3.90	2.24
-48	2-1/2	8	5.17	7.56	3.78	6.00	.81	.94	4.50	2.71
-48	3	8	5.17	8.12	4.50	6.56	.81	.94	4.50	3.31
-64	3-1/2	9	5.54	8.69	5.22	7.12	.81	1.00	5.44	3.81
-64	4	9	5.54	9.25	5.22	7.69	.81	1.00	5.44	4.28

NOTES:

1. Dimensions are in inches.
2. Flange shall be designed to swivel through 360 degrees and dimensioned to mate with 400 lb/in^2 plain flange to MIL-F-20042.
3. Hose nipple material shall be bronze to MIL-B-24480.
4. Socket assembly may be the multiple segment type, bolt or ring retained, so long as envelope dimensions are not exceeded. Material shall conform to QQ-N-281, class A or B or to QQ-N-286, or bronze to MIL-B-24480.
5. Flange material shall be bronze to QQ-C-390, alloy number C90300.

FIGURE 1. Type AI, 400 lb/in^2 flange to hose, straight.

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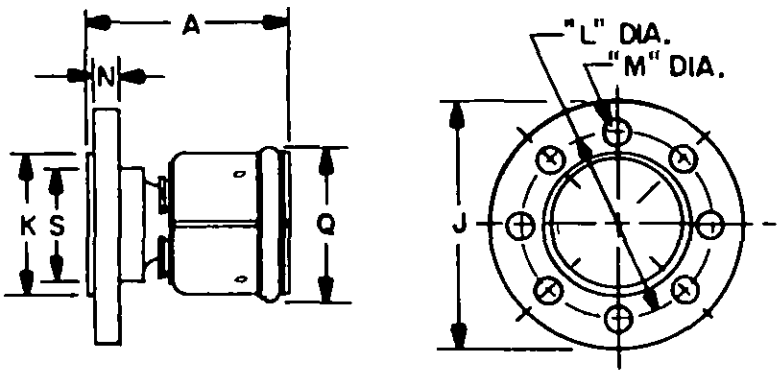
Hose size (dash no.)	Nps	Number of bolt holes	A	J	K	L (bolt circle)	M (bolt holes)	N	Q	S
-40	2	6	5.16	5.56	3.28	4.44	0.69	0.81	3.90	2.24
-40	2	6	5.16	6.12	3.28	5.00	.69	.94	3.90	2.24
-48	2-1/2	6	5.17	6.12	3.78	5.00	.69	.94	4.50	2.71
-48	3	8	5.17	6.62	4.50	5.50	.69	.94	4.50	3.31
-64	3-1/2	8	5.54	7.19	5.22	6.06	.69	1.00	5.44	3.81
-64	4	8	5.54	7.69	5.22	6.06	.69	1.00	5.44	4.28

NOTES:

1. Dimensions are in inches.
2. Hose nipple material shall be bronze to MIL-B-24480.
3. Socket assembly may be the multiple segment type, bolt or ring retained, so long as envelope dimensions are not exceeded. Material may be bronze to MIL-B-24480 or conform to QQ-N-281, class A or B or QQ-N-286.
4. Flange material shall be bronze to QQ-C-390, alloy number C90300.
5. Flange shall be designed to swivel through 360 degrees and dimensioned to mate with 250 lb/in^2 plain flange to MIL-F-20042.

FIGURE 2. Type AII, 250 lb/in^2 flange to hose, straight.

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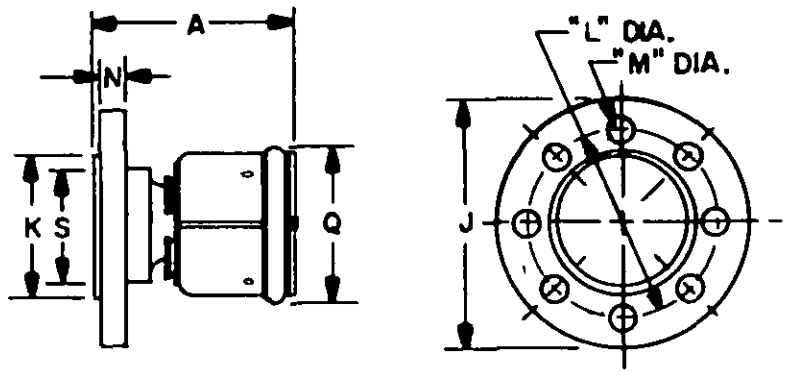
Hose size (dash no.)	Nps	Number of bolt holes	A	J	K	L (bolt circle)	M (bolt holes)	N	Q	S
-40	2	6	5.16	5.56	3.28	4.44	0.56	0.44	3.90	2.24
-40	2-1/2	6	5.16	6.12	3.28	5.00	.56	.50	3.90	2.24
-48	2-1/2	6	5.17	6.12	3.78	5.00	.56	.50	4.50	2.71
-48	3	8	5.17	6.62	4.50	5.50	.56	.50	4.50	3.31
-64	3-1/2	8	5.54	7.19	5.22	6.06	.56	.50	5.44	3.81
-64	4	8	5.54	7.69	5.22	6.56	.56	.50	5.44	4.28

NOTES:

1. Dimensions are in inches.
2. Hose nipple material shall be bronze to MIL-B-24480.
3. Socket assembly may be the multiple segment type, bolt or ring retained, so long as envelope dimensions are not exceeded. Material may be bronze to MIL-B-24480 or conform to QQ-N-281, class A or B or QQ-N-286.
4. Flange material shall be bronze to QQ-C-390, alloy number C90300.
5. Flange shall be designed to swivel through 360 degrees and dimensioned to mate with 150 lb/in^2 plain flange to MIL-F-20042.

FIGURE 3. Type AIII, 150 lb/in^2 flange to hose, straight.

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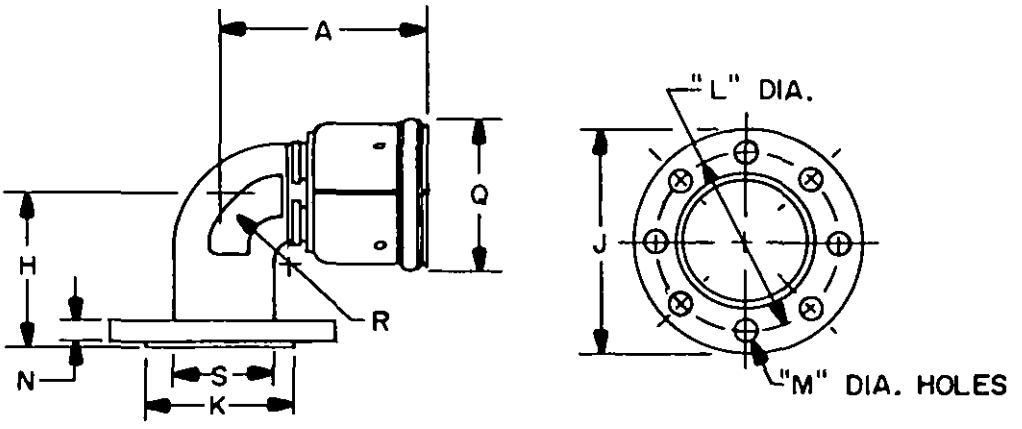
Hose size (dash no.)	Nps	Number of bolt holes	A	J	K	L (bolt circle)	M (bolt holes)	N	Q	S
-40	2	4	5.16	6.00	3.28	4.75	0.75	0.44	3.90	2.24
-40	2-1/2	4	5.16	7.00	3.28	5.50	.75	.50	3.90	2.24
-48	2-1/2	4	5.17	7.00	3.78	5.50	.75	.50	4.50	2.71
-48	3	4	5.17	7.50	4.50	6.00	.75	.50	4.50	3.31
-64	3-1/2	8	5.54	8.50	5.22	7.00	.75	.50	5.44	3.81
-64	4	8	5.54	9.00	5.22	7.50	.75	.50	5.44	4.28

NOTES:

1. Dimensions are in inches.
2. Hose nipple material shall be bronze to MIL-B-24480.
3. Socket assembly may be the multiple segment type, bolt or ring retained, so long as envelope dimensions are not exceeded. Material may be bronze to MIL-B-24480 or conform to QQ-N-281, class A or B or QQ-N-286.
4. Flange material shall be bronze to QQ-C-390, alloy number C90300.
5. Flange shall be designed to swivel through 360 degrees and dimensioned to mate with 150 lb/in^2 plain flange to MIL-F-20042.

FIGURE 4. Type AIV, 150 lb/in^2 flange to hose, straight.

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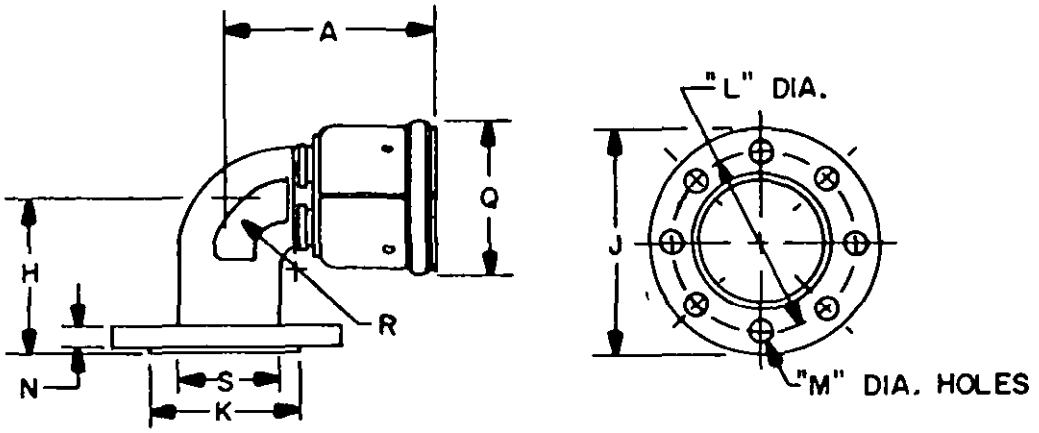
Hose size (dash no.)	Nps	Number of bolt holes	A	H	J	K	L (bolt circle)	M (bolt holes)	N	Q	R	S
-40	2	7	5.70	3.50	6.50	3.28	5.19	0.69	0.81	3.90	1.88	2.24
-48	2-1/2	8	5.96	4.00	7.56	3.78	6.00	.81	.94	4.50	2.09	2.71
-48	3	8	5.96	4.00	8.12	4.50	6.56	.81	.94	4.50	2.09	3.31
-64	3-1/2	9	6.75	4.62	8.69	5.22	7.12	.81	1.00	5.44	2.81	3.81
-64	4	9	6.75	4.62	9.25	5.22	7.69	.81	1.00	5.44	2.81	4.28

NOTES:

1. Dimensions are in inches.
2. Flange shall be designed to swivel through 360 degrees and dimensioned to mate with 400 lb/in^2 plain flange to MIL-F-20042.
3. Hose nipple material shall be bronze to MIL-B-24480.
4. Socket assembly may be the multiple segment type, bolt or ring retained, so long as envelope dimensions are not exceeded. Material may be bronze to MIL-B-24480 or conform to QQ-N-281, class A or B or QQ-N-286.
5. Flange material shall be bronze to QQ-C-390, alloy number C90300.

FIGURE 5. Type BI, 400 lb/in^2 flange to hose, 90-degree elbow.

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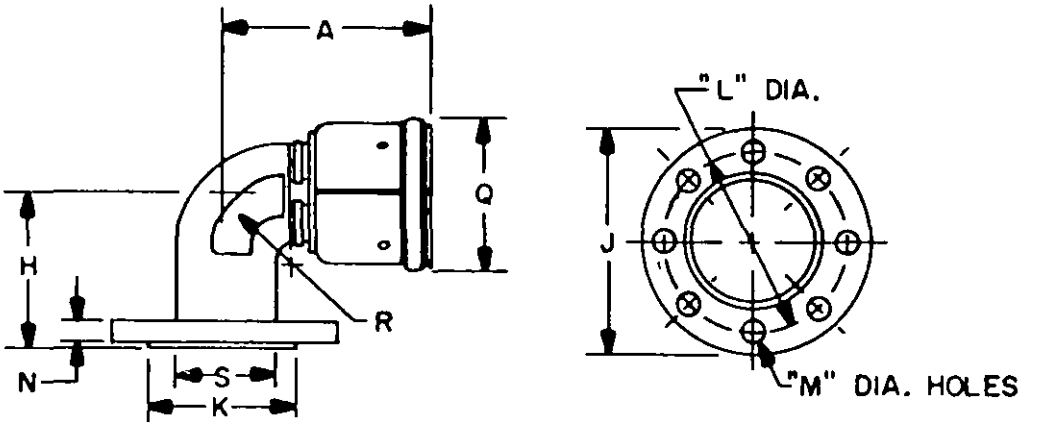
Hose size (dash no.)	Nps	Number of bolt holes	A	H	J	K	L (bolt circle)	M (bolt holes)	N	Q	R	S
-40	2	6	5.70	3.50	5.56	3.28	4.44	0.69	0.81	3.90	1.88	2.24
-40	2-1/2	6	5.70	3.50	6.12	3.28	5.00	.69	.94	3.90	1.88	2.24
-48	2-1/2	6	5.96	4.00	6.12	3.78	5.00	.69	.94	4.50	2.09	2.71
-48	3	8	5.96	4.00	6.62	4.50	5.50	.69	.94	4.50	2.09	3.31
-64	3-1/2	8	6.75	4.62	7.19	5.22	6.06	.69	1.00	5.44	2.81	3.81
-64	4	8	6.75	4.62	7.69	5.22	6.56	.69	1.00	5.44	2.81	4.28

NOTES:

1. Dimensions are in inches.
2. Hose nipple material shall be bronze to MIL-B-24480.
3. Socket assembly may be the multiple segment type, bolt or ring retained, so long as envelope dimensions are not exceeded. Material may be bronze to MIL-B-24480 or conform to QQ-N-281, class A or B or QQ-N-286.
4. Flange material shall be bronze to QQ-C-390, alloy number C90300.
5. Flange shall be designed to swivel through 360 degrees and dimensioned to mate with 250 lb/in^2 plain flange to MIL-F-20042.

FIGURE 6. Type BII, 250 lb/in^2 flange to hose, 90-degree elbow.

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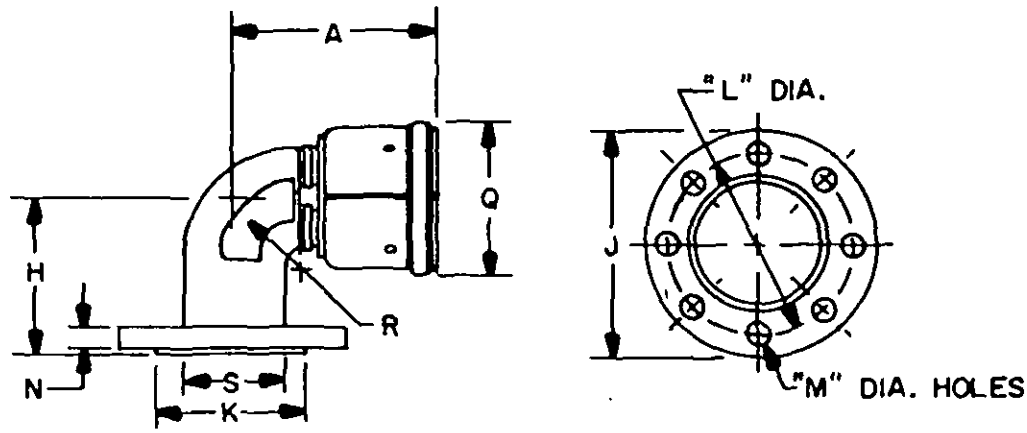
Hose size (dash no.)	Nps	Number of bolt holes	A	H	J	K	L (bolt circle)	M (bolt holes)	N	Q	R	S
-40	2	6	5.70	3.50	5.56	3.28	4.44	0.56	0.44	3.90	1.88	2.24
-40	2-1/2	6	5.70	3.50	6.12	3.28	5.00	.56	.50	3.90	1.88	2.24
-48	2-1/2	6	5.96	4.00	6.12	3.78	5.00	.56	.50	4.50	2.09	2.71
-48	3	8	5.96	4.00	6.62	4.50	5.50	.56	.50	4.50	2.09	3.31
-64	3-1/2	8	6.75	4.62	7.19	5.22	6.06	.56	.50	5.44	2.81	3.81
-64	4	8	6.75	4.62	7.69	5.22	6.56	.56	.50	5.44	2.81	4.28

NOTES:

1. Dimensions are in inches.
2. Hose nipple material shall be bronze to MIL-B-24480.
3. Socket assembly may be the multiple segment type, bolt or ring retained, so long as envelope dimensions are not exceeded. Material may be bronze to MIL-B-24480 or conform to QQ-N-281, class A or B or QQ-N-286.
4. Flange material shall be bronze to QQ-C-390, alloy number C90300.
5. Flange shall be designed to swivel through 360 degrees and dimensioned to mate with 150 lb/in^2 plain flange to MIL-F-20042.

FIGURE 7. Type BIII, 150 lb/in^2 flange to hose, 90-degree elbow.

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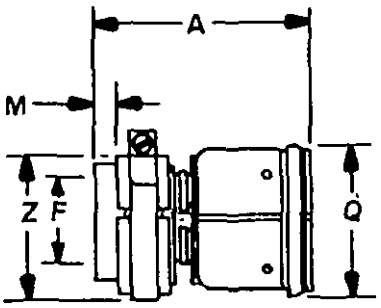
Hose size (dash no.)	Nps	Number of bolt holes	A	H	J	K	L (bolt circle)	M (bolt holes)	N	Q	R	S
-40	2	4	5.70	3.50	6.00	3.28	4.75	0.75	0.44	3.90	1.88	2.24
-40	2-1/2	4	5.70	3.50	7.00	3.28	5.50	.75	.50	3.90	1.88	2.24
-48	2-1/2	4	5.96	4.00	7.00	3.78	5.50	.75	.50	4.50	2.09	2.71
-48	3	4	5.96	4.00	7.50	4.50	6.00	.75	.50	4.50	2.09	3.31
-64	3-1/2	8	6.75	4.62	8.50	5.22	7.00	.75	.50	5.44	2.81	3.81
-64	4	8	6.75	4.62	9.00	5.22	7.50	.75	.50	5.44	2.81	4.28

NOTES:

1. Dimensions are in inches.
2. Hose nipple material shall be bronze to MIL-B-24480.
3. Socket assembly may be the multiple segment type, bolt or ring retained, so long as envelope dimensions are not exceeded. Material may be bronze to MIL-B-24480 or conform to QQ-N-281, class A or B or QQ-N-286.
4. Flange material shall be bronze to QQ-C-390, alloy number C90300.
5. Flange shall be designed to swivel through 360 degrees and dimensioned to mate with 150 lb/in^2 plain flange to MIL-F-20042.

FIGURE 8. Type BIV, 150 lb/in^2 flange to hose, 90-degree elbow.

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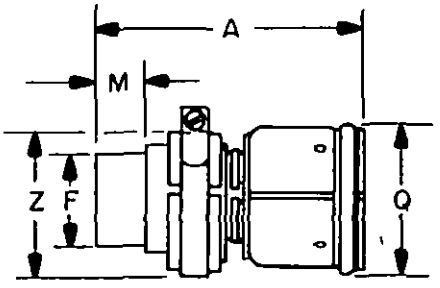
SH 1320239

Hose size (dash no.)	Nps	A	F	M	Q	Z
-40	2	6.10	2.38	0.66	3.90	4.00
-48	2-1/2	6.42	2.88	.78	4.50	4.42
-48	3	6.11	3.50	.83	4.50	5.19
-64	3-1/2	6.16	4.00	.88	5.44	5.56
-64	4	6.31	4.50	.91	5.44	6.12

NOTES:
1. Dimensions are in inches.
2. Hose nipple material shall be bronze to MIL-B-24480.
3. Socket assembly may be the multiple segment type, bolt or ring retained, so long as envelope dimensions are not exceeded. Material shall be bronze to MIL-B-24480 or conform to QQ-N-281, class A or B or QQ-N-286.
4. Split clamp assembly shall be to figure 15 as applicable.
5. Silver braze tailpiece shall be to figure 16 as applicable.

FIGURE 9. Type C, split clamp to hose, straight.

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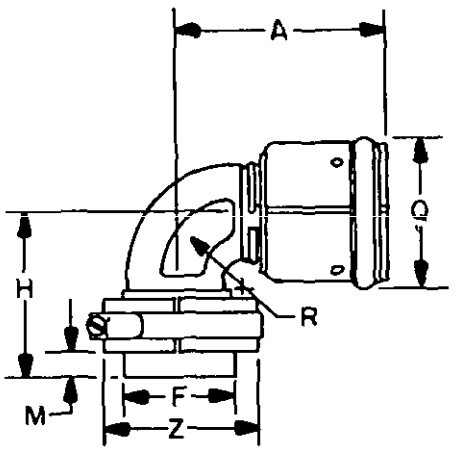
SH 1320240

Hose size (dash no.)	Nps	A	F	M	Q	Z
-40	2	6.61	2.38	1.31	3.90	4.00
-48	2-1/2	6.60	2.88	1.12	4.50	4.42
-48	3	8.01	3.50	2.50	4.50	5.19
-64	3-1/2	8.02	4.00	2.50	5.44	5.56
-64	4	8.38	4.50	2.64	5.44	6.12

NOTES:
1. Dimensions are in inches.
2. Hose nipple material shall be bronze to MIL-B-24480.
3. Socket assembly may be the multiple segment type, bolt or ring retained, so long as envelope dimensions are not exceeded. Material may conform to QQ-N-281, class A or B or to QQ-N-286, or bronze to MIL-B-24480.
4. Split clamp assembly shall be to figure 15 as applicable.
5. Silver braze tailpiece shall be to figure 16 as applicable.

FIGURE 10. Type CA, split clamp to hose, straight.

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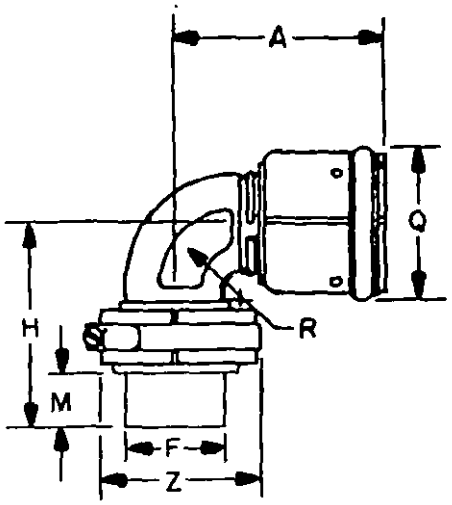
Hose size (dash no.)	Nps	A	F	H	M	Q	R	Z
-40	2	5.70	2.38	3.88	0.66	3.90	1.88	4.00
-48	2-1/2	5.88	2.88	4.51	.78	4.50	2.09	4.42
-48	3	6.46	3.50	4.25	.83	4.50	2.09	5.19
-64	3-1/2	6.75	4.00	5.14	.88	5.44	2.81	5.56
-64	4	6.75	4.50	5.27	.91	5.44	2.81	6.12

NOTES:

1. Dimensions are in inches.
2. Hose nipple material shall be bronze to MIL-B-24480.
3. Socket assembly may be the multiple segment type, bolt or ring retained, so long as envelope dimensions are not exceeded. Material shall be bronze to MIL-B-24480 or conform to QQ-N-281, class A or B or to QQ-N-286.
4. Split clamp assembly shall be to figure 15 as applicable.
5. Silver braze tailpiece shall be to figure 16 as applicable.

FIGURE 11. Type D, split clamp to hose, 90-degree elbow.

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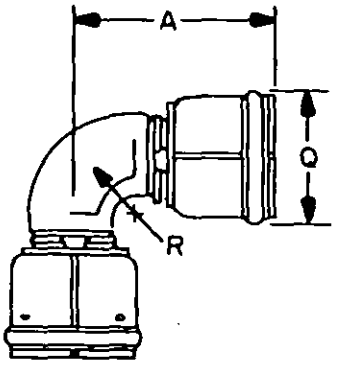
Hose size (dash no.)	Nps	A	F	H	M	Q	R	Z
-40	2	5.70	2.38	4.76	1.31	3.90	1.88	4.00
-48	2-1/2	5.88	2.88	4.71	1.12	4.50	2.09	4.42
-48	3	6.46	3.50	6.15	2.50	4.50	2.09	5.19
-64	3-1/2	6.75	4.00	6.99	2.50	5.44	2.81	5.56
-64	4	6.75	4.50	7.24	2.64	5.44	2.81	6.12

NOTES:

1. Dimensions are in inches.
2. Hose nipple material shall be bronze to MIL-B-24480.
3. Socket assembly may be the multiple segment type, bolt or ring retained, so long as envelope dimensions are not exceeded. Material may be bronze to MIL-B-24480 or conform to QQ-N-281, class A or B or to QQ-N-286.
4. Split clamp assembly shall be to figure 15 as applicable.
5. Silver braze tailpiece shall be to figure 17 as applicable.

FIGURE 12. Type DA, split clamp to hose, 90-degree elbow.

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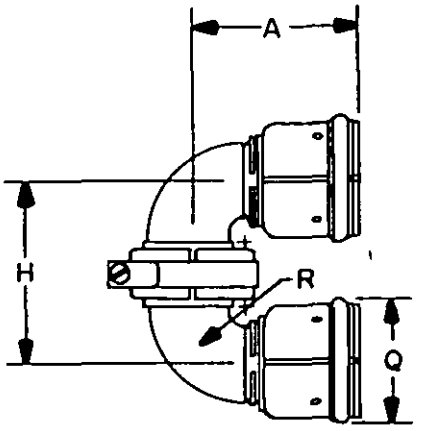
SH 1320243

Hose size (dash no.)	A	Q	R
-40	5.70	3.90	1.88
-48	5.96	4.50	2.09
-64	6.75	5.44	2.81

NOTES:
1. Dimensions are in inches.
2. Hose nipple material shall be bronze to MIL-B-24480.
3. Socket assembly may be the multiple segment type, bolt or ring retained, so long as envelope dimensions are not exceeded. Material may be bronze to MIL-B-24480 or conform to QQ-N-281, class A or B or to QQ-N-286.

FIGURE 13. Type E, hose to hose, 90-degree elbow.

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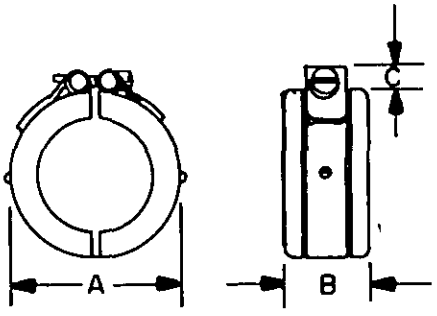
SH 1320244

Hose size (dash no.)	Nps	A	H	Q	R
-40	2	5.70	5.40	3.90	1.88
-48	2-1/2	6.05	5.92	4.50	2.09
-48	3	6.46	5.95	4.50	2.09
-64	3-1/2	6.75	7.32	5.44	2.81
-64	4	6.75	7.44	5.44	2.81

NOTES:
1. Dimensions are in inches.
2. Hose nipple material shall be bronze to MIL-B-24480.
3. Socket assembly may be the multiple segment type, bolt or ring retained, so long as envelope dimensions are not exceeded. Material may be bronze to MIL-B-24480 or conform to QQ-N-281, class A or B or QQ-N-286.
4. Split clamp assembly shall be to figure 15 as applicable.

FIGURE 14. Type F, split clamp fitting, 180-degree elbow.

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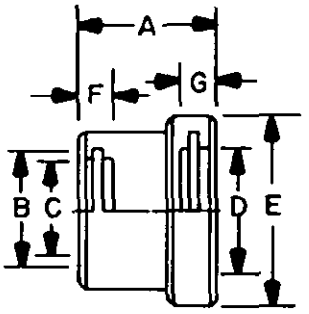
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Nps	A	B	C
2	4.00	1.38	0.39
2-1/2	4.42	1.44	.39
3	5.19	1.50	.39
3-1/2	5.56	1.50	.39
4	6.12	1.62	.39

NOTES:
1. Dimensions are in inches.
2. Split clamp halves shall be bronze to QQ-C-390, alloy number C95500.
3. Screws and nuts shall be bronze or conforming to QQ-C-591, alloy number 651 or 655, or QQ-N-281.
4. Band shall be to QQ-N-281.

FIGURE 15. Split clamp assembly.

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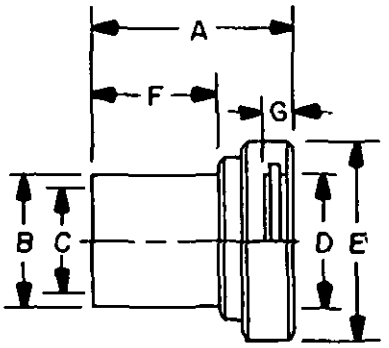
SH 1320246

Nps	A	B	C	D	E	F	G
2	1.64	2.38	2.12	2.51	3.25	0.66	0.35
2-1/2	1.86	2.88	2.59	3.01	3.75	.78	.35
3	1.94	3.50	3.17	3.64	4.36	.83	.35
3-1/2	1.98	4.00	3.63	4.14	4.93	.88	.35
4	2.10	4.50	4.11	4.64	5.43	.91	.42

NOTES:
1. Dimensions are in inches.
2. Material for silver braze tailpiece shall be bronze to QQ-C-390, alloy number C90300, or MIL-B-24480.
3. Braze socket and groove shall be designed for ultrasonic inspection.
4. Back-up ring shall be commercial grade leather.
5. O-ring shall be synthetic rubber suitable for system service.

FIGURE 16. Female silver braze tailpiece.

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Nps	A	B	C	D	E	F	G
2	2.17	2.38	2.04	2.51	3.24	1.31	0.35
2-1/2	2.04	2.88	2.59	3.01	3.74	1.12	.35
3	3.45	3.50	3.00	3.63	4.37	2.50	.35
3-1/2	3.45	4.00	3.50	4.13	4.93	2.50	.35
4	3.67	4.50	3.88	4.63	5.43	2.64	.42

NOTES:
1. Dimensions are in inches.
2. Material for silver braze tailpiece shall be bronze to QQ-C-390, alloy number C90300, or MIL-B-24480.
3. Back-up ring shall be commercial grade leather.

FIGURE 16. Female silver braze tailpiece.

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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